**SET - A**

| Register No | | | | | | | | | | | | | | | |
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**SRM INSTITUTE OF SCIENCE AND TECHNOLOGY**
**21CSC555J - MACHINE LEARNING ALGORITHM**
**CYCLE TEST – I**

**Program Offered: M. Tech-AI&DS** | **Year / Semester: I / I**
**Max. Marks: 50** | **Duration: 2hr 30 Mins**
**Course Code: 21CSC555J** | **Date of Exam: 26/10/2024**

**Part A (2 x 20 = 40 marks)**

**Answer any two questions**

1. (i). Explain simple linear regression using an example dataset, along with a detailed breakdown of the mathematical calculations involved in deriving the equation? **10**

   (ii)What are the evaluation metrics that are used for regression in Machine Learning? explain Mean Absolute Error (MAE) and Mean Squared Error (MSE)? **5**

   (iii). Explain the concept of Overfitting and underfitting? **5**

2 (i). What is Logistic Regression? How you will convert the line equation into Sigmoid curve? **5**

   (ii). How you will handle more than two class using Logistic regression? Explain with example? **10**

   (iii). What is Regularization and explain different types of it? **5**

3 (i). Explain in detail about Cross validation and its types? **10**

   (ii). What are the different performance metrics of a Classification model? Explain Confusion matrix table with example? **5**

   (iii). What is loss function, explain how Gradient descent helps to Minimize the loss **5**

**Practicals**

1 How can we use simple linear regression to predict an individual's salary based on their years of experience? What are the key steps involved in building and evaluating this model? **20**